

Safety and privacy in the cloud

Anyone who works with data inherits the responsibility to protect private or personal data, especially **personally identifiable information, or PII**.

This is all about data privacy, which we'll explore in this video.

PII and other privacy standards have legal implications.

So, it's important to call out: Our discussion in this course should not be considered legal advice.

So, what is data privacy?

Data privacy is preserving a data subject's information any time a data transaction occurs.

Any organization that collects data knows that earning users' trust is an essential part of working with their data.

As a cloud data professional, one of your primary goals will be to use **data honestly, transparently, and fairly**.

PII is data that is unique to an individual and therefore can be used to distinguish one person from another.

A user's email address, mailing address, phone number, precise location, birthday, and full name are all examples of PII —and must be safeguarded.

Along those same lines, **personal health data is PHI, or protected health information.**

PHI is health data that can identify an individual, like information about **patient demographics, mental or physical health diagnoses or treatments, and payment records related to health care.**

It's also important to know about the **General Data Protection Regulations, or GDPR.**

This is **privacy legislation** for organizations in **Europe**.

It regulates how they can collect, use, and store personal data.

It also has requirements around reporting to increase accountability and fines for those that fail to comply.

A pro tip is that, even if your employer isn't located in Europe, the rules may apply to any business that collects data from European citizens.

Whether complying with GDPR or trying to follow best practices in PII and PHI data security, there are several ways to help your organization avoid fraud and identity theft.

One strategy is **identity access management**.

This is a process that gives each employee who interacts with the company's system access to specifically defined programs and datasets.

As a data professional, you may be required to only access the bare minimum of data needed to do your job.

This is usually referred to as **NTK, or "need to know."**

Accessing data just out of curiosity or for a non-valid business reason may be **prohibited**.

Another proven strategy is having internal data stewards.

This privacy team is in charge of keeping data access in check.

Frequent audits within your organization are an additional security measure.

An audit is a formal examination of how users are accessing data in order to ensure safe and appropriate access while identifying and solving any data concerns.

Another way to protect data is through the use of **security keys** for accessing software, accounts, and datasets.

This is an **authentication method that uses a physical digital signature, or key, to verify a user's identity before they access these resources.**

Finally, **encryption** is another proven method.

This is just the process of encoding information.

Encryption protects data by scrambling it into a unique secret code that can only be unlocked with a digital key.

Data may be encrypted on an individual machine and unencrypted when you work with it.

Or it may be encrypted when it's shared or sent from one machine to another.

Having access to data —especially data about people— is a big responsibility!

So it's really important for cloud data professionals to know the best practices of data privacy and safety.